

CHE 330 Week 3 Homework

Assigned 9/10/2025, Due 9/16/2025 11:59 PM

1. Compare and contrast 3 methods of polymerization.

	Comparison	Contrast
Bulk	Simplest method, no solvent or dispersant	High reaction rates and molecular weights but heat removal is difficult
Solution	Monomers are dissolved in a solvent and the polymer remains in solution	Facilitates heat transfer and control leads to cleaner products, but has slower reaction compared to bulk
Emulsion	Uses surfactants and water to create an emulsion of monomer droplets	Polymerization occurs within the droplets offering excellent control over heat and reaction rates, but often results in lower molecular weight polymers.

✓ 3

2. Highlight key differences between step and chain polymerization.

Step

- slow MW build-up
- Long reaction time needed for high MW
- No termination end groups stay reactive

Chain

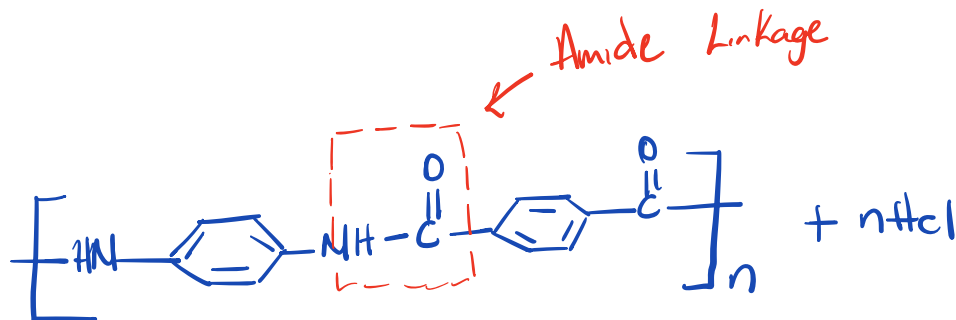
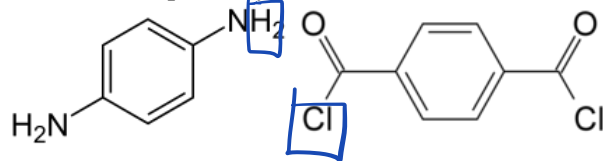
- fast → high MW formed early
- high MW not dependent on time
- Requires initiation, propagation, termination

✓ 2

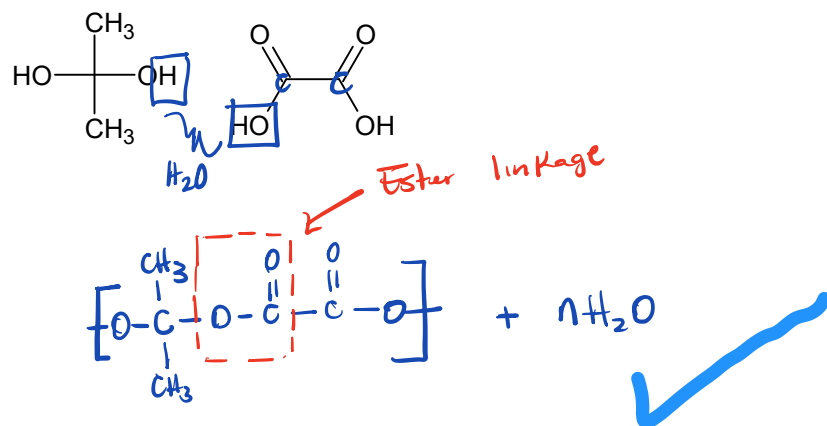
3. Identify the differences in the polymer structure, starting materials and method of formation of **Nylon 6** and **Nylon 6,6** that are produced commercially.

Nylon 6	Nylon 6,6
Polymer Structure : $[NH-(CH_2)_5-CO]_n$	$[NH(CH_2)_6NHCO(CH_2)_4CO]_n$ ✓
Starting Material : Caprolactam	hexamethylenediamine and adipic acid ✓
Method of Formation : Formed through ring-opening polymerization of Caprolactam	Formed through condensation polymerization of hexamethylenediamine and adipic acid, with the elimination of water molecules ✓

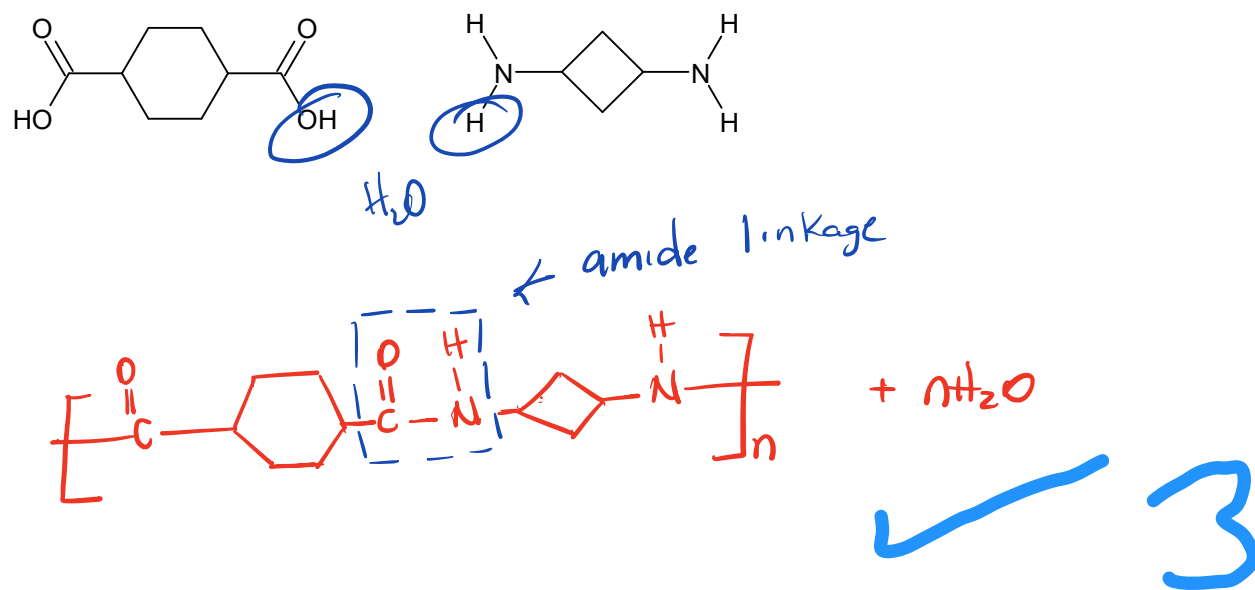
4. For each of the following monomer/starting materials, draw the polymer structure formed
- p-Phenylenediamine and Terephthaloyl chloride



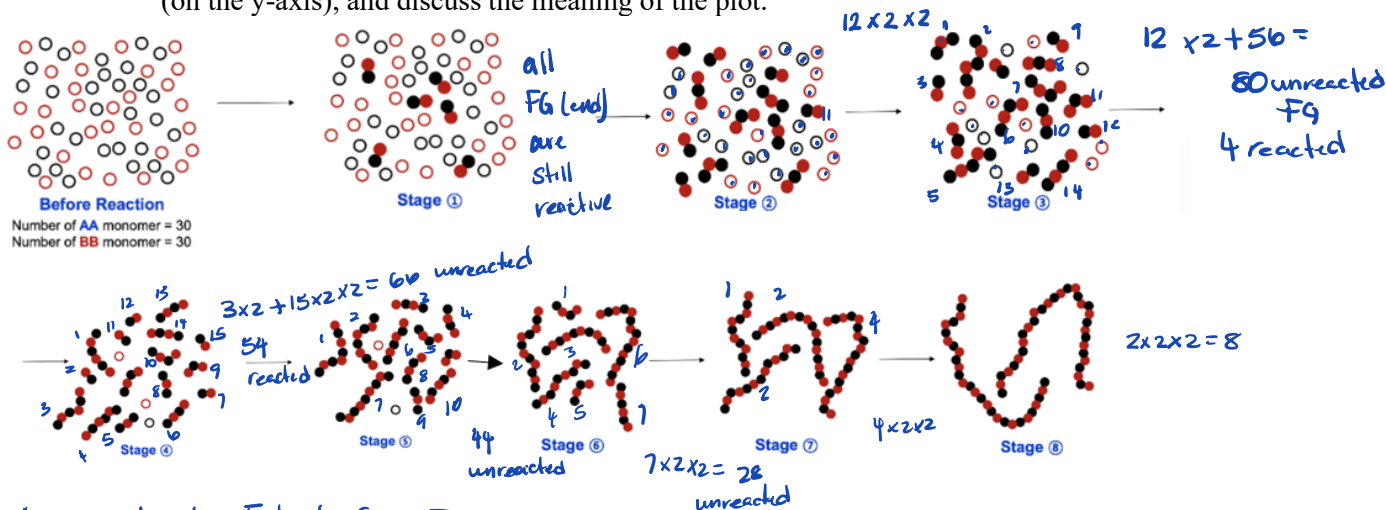
Propane-2,2-diol and oxalic acid



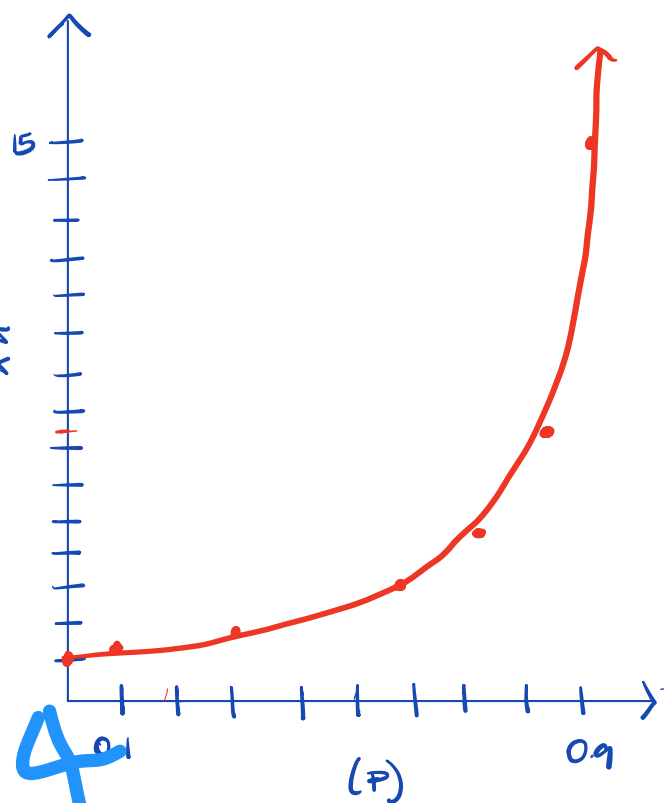
cyclohexane-1,4-dicarboxylic acid and cyclobutane-1,3-diamine



5. Consider the step polymerization of 60 bifunctional molecules shown below.
- Compute the number of functional groups at each stage of the reaction.
 - Determine the extent of the reaction and degree of polymerization at each stage of the reaction.
 - Make a plot of the extent of reaction, p (on the x-axis) vs. degree of polymerization, $\bar{x}_n$ (on the y-axis), and discuss the meaning of the plot.

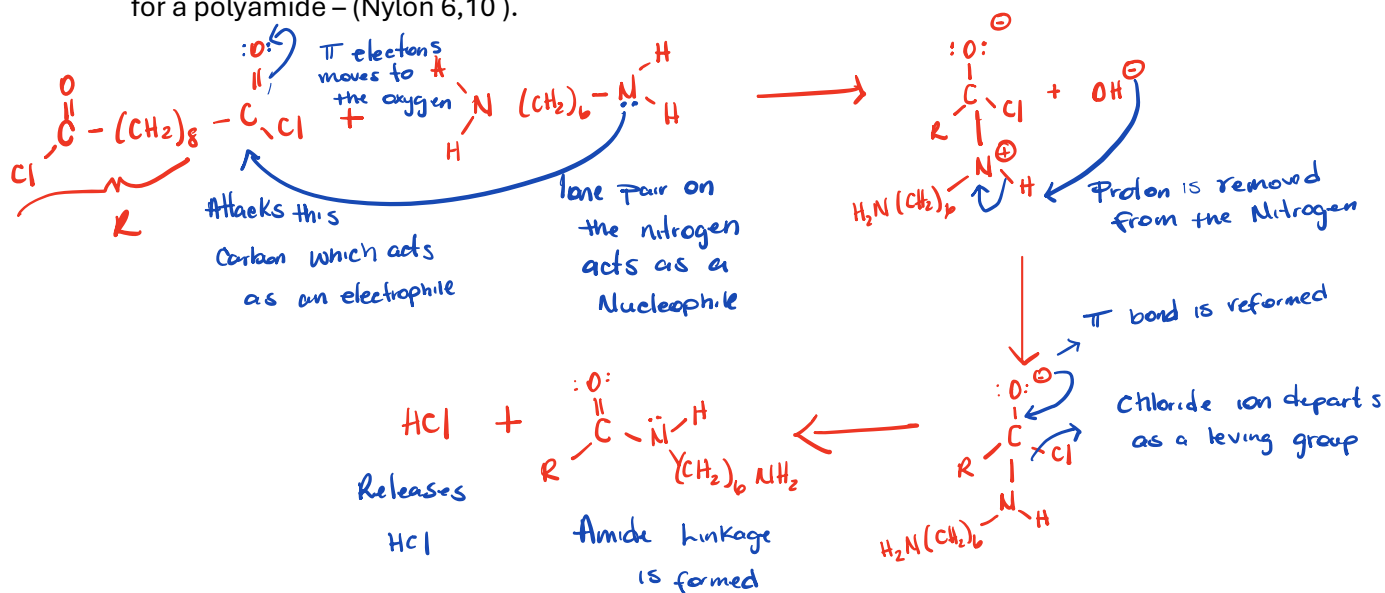


Stage	# of functional Groups	Extent of Reaction (p)	Degree of polymerization ($\bar{x}_n$)
Before	120	$0/120 = 0$	$120/120 = 1$
1	120	0	1
1 2	12 reacted FG 108 unreacted FG	$\frac{12}{120} = 0.1$	$\frac{120}{108} = 1.11$ ✓
2 3	40 reacted FG 80 unreacted FG	$\frac{40}{120} = 0.33$	$\frac{120}{80} = 1.5$ ✗
3 4	54 reacted FG 66 unreacted FG	$\frac{54}{120} = 0.45$	$\frac{120}{66} = 1.82$ ✗
4 5	76 reacted FG 44 unreacted FG	$\frac{76}{120} = 0.63$	$\frac{120}{44} = 2.73$ ✗
5 6	92 reacted FG 28 unreacted FG	$\frac{92}{120} = 0.76$	$\frac{120}{28} = 4.29$ ✗
6 7	104 reacted FG 16 unreacted FG	$\frac{104}{120} = 0.87$	$\frac{120}{16} = 7.5$ ✗
7 8	112 reacted FG 8 unreacted FG	$\frac{112}{120} = 0.93$	$\frac{120}{8} = 15$ ✓



As the reaction approaches completion ($p = 1$), a small increase in the extent of reaction results in a significant increase in the average chain length.

6. Using arrow diagrams, clearly explain and illustrate the mechanism of step polymerization for a polyamide – (Nylon 6,10).



✓
10